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Professional Profile

Senior Software Engineer with 14+ years of experience in advanced computing technologies and full-stack tech-solutions development.
Proven experience in real-time ad-serving systems, globally replicated distributed data-store systems, live music/audio broadcast applications, and location-based & data-mobilization software solutions.
Currently tech lead of the core filtering services of Amazon Ads' next-generation e-commerce retail ad server; leads designs spanning nine-plus teams and launches that roll out worldwide.

Experience

- **Amazon.com, Inc.** Jan 2020 - Present
  Senior Software Development Engineer (L6) *SDE II, Jan 2020 - Sep 2025*
  - **Next-Generation Ad Server**: The core ad-delivery engine of Amazon Ads' next-generation e-commerce ad server, serving Sponsored Products and Brands across Amazon retail and third-party supply, which lets merchants and sellers promote featured placements of their products. On every shopper request it decides, within a **200 ms p99** latency budget, which candidate ads may be shown, applying advertiser targeting, the shopper's search filters (brand, price, delivery speed), compliance, and budget rules. Built on AWS ECS with a **terabyte-scale** distributed ElastiCache.
    – **Tech lead**, leading 2–5 engineers and acting as design-review and operational gatekeeper for the service.
    – **Sole-authored the design** for enforcing shopper search filters natively in the new ad server, aligned **four organizations** behind it, and shipped the resulting filter chain now serving Search, Product Detail Page, Browse, and Agentic Browse.
    – Led the migration of product detail-page ad filtering off the incumbent ad-serving engine, Fenix, with placement-by-placement verification that the new service matched the old.
    – Led the design and, with the team, built a rules-driven engine for the business rules of themed ad widgets, now live on product detail, cart, and checkout pages.
    – Owned production reliability and readiness: traffic-invariant alarms, availability-zone-aligned caches, fail-closed health checks, and a **3x** capacity increase for the busiest cell; led the region-by-region recovery from a production incident caused by a network-configuration change.
  - **Ads on Filtered Search Results**: Shoppers narrow search results with filters such as brand, price, or delivery speed; Sponsored Products had been serving those result pages without ads. Cross-organization program between Sponsored Ads and Retail Search, which owns search across Amazon's retail catalog, to close that gap.
    – Owned the joint goal; prototyped the integration with the Retail Search interface and load-tested it at production request rates.
    – Designed and led the auction-stage integration experiment across the ad-serving, auction, and Search teams, then rolled it out worldwide: **+0.3% Sponsored Products revenue, about $88MM projected annual**, from roughly **60K net-new ad clicks per day**.
    – Made new search filters supportable **on launch day instead of after 12+ weeks** of bespoke engineering each.
    – Delivered the Brand filter (**+0.4%, about $6.3MM projected annual**) after root-causing a thread-safety defect in Fenix's filtering framework.
    – Drove the Prime same-day-delivery filter through staged rollout to **all US traffic and 22 countries**: **about $22.4MM projected annual revenue**, with filter latency cut from **2 ms to 1 ms**.
    – When a tail-latency regression threatened a rollback, proved through a multi-week A/B analysis that it was confined to a small slice of traffic and won an organization-level exemption.
    – Presented the approach at the ad-server all-hands; since then act as the **standing technical authority**, consulted by **four partner teams** on their own filter launches.

- **Global Ad Campaigns**: Led the cross-team design (**nine-plus teams**, two Principal Engineers, a Principal TPM) that enabled global ad campaigns, one campaign serving across countries, for **1.08M+ advertisers**, by making campaign-eligibility decisions available across regions with no latency regression.
  – Designed the cross-region path: a connectivity layer plus a reactive cache on DynamoDB Global Tables, sized from 48M+ requests and 200 GB of production logs across three regions.
  – The cache reached a **54–62% hit rate** at a one-day TTL and cut cross-region latency by **about 90 ms**.
  – Revived the load-testing framework and wrote the runbooks the team still uses for launch readiness; unblocked five upstream services and the Sponsored Products global-campaigns launch.
- **Amazon Music Amp**: Amp is an infinite-dial digital radio broadcast mobile application. Amp allows users to create a show, go live, play music, and take in call-in guests all with simple flow and within minutes.
  – Lead engineer developing Amp Android app and migrating Amp iOS app to hybrid ReactNative-based architecture with native core components in both iOS Swift and Android Kotlin.
  – Team leader of Creator Studio & Call-in experience full-stack development maintaining iOS/Swift client and backend system using AWS Lambda, Chime APIs, Elastic-Containers on Fargate microservices, and AWS AppSync GraphQL.
  – Led two engineers and a cross-organization effort with the AWS Chime team to develop the iOS Bluetooth/audio solution that lets creators talk over music on live broadcasts within music-licensing rules; filed as a **US patent application**, `SWITCHING WIRELESS PROFILES IN RESPONSE TO TRIGGER EVENTS`.
- **Ads Serving Core Engine**: `FenixSPService` (a.k.a. Fenix) is orchestration engine that serves Sponsored-Products Ads on Amazon.com retail website and mobile app worldwide. Fenix is a high-throughput low-latency real-time distributed service that delivers curated ads per user.
  – Worked on migration to Native AWS (NAWS) delivering the first proof-of-concept of Fenix on the public/native offering of AWS ecosystem.
  – Identified **+20 technical blockers**, led team of engineers to develop simple workaround solutions, and eventually provided a risk-mitigation path for full migration and launch of the service.
  – Launching Fenix on Native AWS was later featured in AWS re:Invent 2021 - Under the hood at Amazon Ads.
  – Authored the remote-debugging runbook for Fenix on Native AWS, still used by **400+ engineers** across the ad-server organization and partner teams.
  – Participated in on-call and DevOps rotations; improving pipelines and monitoring to implement best practices of Continuous Integration & Delivery CI/CD.
- **Engineering Excellence & Mentorship**: Mechanisms and people work across the ad-serving organization.
  – Reviewed **148 code changes for 25 engineers across six-plus organizations** in one half, while authoring **67 changes across 17 packages**.
  – Stood up engineering-governance mechanisms for the new ad server's core filtering services (canonical runbooks, an automated code-review guardrail, reusable knowledge-base skills), since adopted by **three partner organizations**.
  – Built developer-velocity tooling with agentic GenAI coding tools; moved all eight of the service's packages to JDK 21.
  – Mentored five engineers, onboarded three new hires, wrote promotion feedback for five, and conducted seven interview loops; in training as a change-management Bar Raiser.

- **Webalo, Inc.** Jul 2016 - Dec 2019
  Software Engineer
  - **HTML Toolkit**: Designed the architecture of a platform-agnostic, HTML-based UI widget-rendering toolkit and led the team that migrated Webalo's User Agent applications onto it, retiring the Java Canvas-based toolkit (TypeScript, Web Components, Angular, Bootstrap).
  - **CI/CD**: Automated the system build, integration and end-to-end test execution, and binary distribution with Git Hooks, Jenkins, and shell scripting.
  - **Unit & Integration Testing**: Set up and designed the system testing framework for integration tests: TestNG and Mockito for module testing, Selenium for end-to-end testing.

- **Google, Inc.** Aug 2015 - Nov 2015
  Software Engineering Intern
  - **Sandlot**: As part of the G-Tech team, designed and built sentiment prediction inside Google's partner-management tool (Python, AngularJS).

- **Wireless Stars, LLC** Oct 2013 - May 2015
  Software Engineer
  - **ARTS**: Location-based road-traffic estimation from cell-phone sensor data, developed side by side with the research team in an R&D setting: built the Android sensor-collection SDK, implemented and validated the Kalman-filter speed estimator (sensor fusion), the Spring Boot and Apache Camel back-end modules, PostGIS spatial queries, and the Android app used for demos and system evaluation.
  - **EYE360**: Augmented reality application with bi-directional user interaction on Unity3D and Qualcomm Vuforia SDK.

Experience Areas & Tools[1]

- **Advanced Computing**: Real-time Ad Serving and Filtering at Scale – Distributed Cloud Computing – Global Geo-separated Data Stores – Distributed Caching – Serverless Architectures
- **Data Protection and Cybersecurity**: DDOS Protection, WAF, and API Authentication Protocols – Cryptology and Data Privacy
- **Tech Business Domains**: E-Commerce and Advertising – Music and Live Broadcast – Digital Data Transformation – Location-based Services – Augment Reality
- **Languages**: Java – Python – Swift – TypeScript – HTML/JavaScript/CSS – SQL
- **Frameworks**: iOS – ReactNative – Spring MVC/Boot – Bootstrap – Angular – Unity3D – Android
- **CI/CD**: CDK– Jenkins – TestNG – Selenium – Mockito/PowerMock – Jasmine/Karma
- **GenAI / Agentic Development**: Amazon Q – agentic coding workflows – LLM-assisted developer tooling

Education

- **Alexandria University, Faculty of Engineering** Sep 2007 – Jul 2012
  B.Sc. Computer and Systems Engineering
- **Alexandria University, Special Scientific Program** Sep 2012 - Jan 2013
  Teaching Assistant: Introduction to Databases, Numerical Analysis

Continuous Learning [2]

- **Agentic AI: Multi-Agent Collaboration**: Amazon Machine Learning University (2026)
- **Generative AI with Large Language Models**: DeepLearning.AI and AWS, Coursera (2025)
- **Architecture and Developing on AWS**: AWS Hands-on Training
- **DevOps on AWS**: AWS Hands-on Training
- **Security Engineering on AWS**: AWS Hands-on Training
- **Programming Languages**: Coursera - Washington Uni
- **Introduction to Functional Programming in Scala**: Coursera
- **Applied Cryptography**: Udacity
- **Introduction to Cryptography**: Coursera - Stanford
- **Introduction to Recommender Systems**: Coursera - Uni of Minnesota
- **Algorithms Design and Analysis**: Coursera - Stanford
- **CSMM.102x Machine Learning**: Edx - ColumbiaX
- **CSMM.101x Artificial Intelligence**: Edx - ColumbiaX
- **Introduction to AI for Robotics**: Udacity
- **Machine Learning**: Coursera
- **Coding the Matrix**: Coursera
- **Scalable Machine Learning**: Edx

[1] Technologies are listed in decreasing order of experience.
[2] Certificates are available upon request.